L-11

23/08/25

Laplace of derivatives:

1. $\mathcal{L}\{y(t)\} = Y(s)$
2. $\mathcal{L}\{y'(t)\} = sY(s) - y(0)$
3. $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$
4. $\mathcal{L}\{y'''(t)\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$
5. $\mathcal{L}\{y^{iv}(t)\} = s^4Y(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)$
6. $\mathcal{L}\{y^n(t)\} = s^nY(s) - s^{n-1}y(0) - s^{n-2}y'(0) - s^{n-3}y''(0) \ldots - y^{(n-1)}(0)$

Initial value problem	Boundary value problem
initial condition	Boundary condition
$y(0) = A$	$y(0) = A$
$y'(0) = B$	$y'(2) = B$
$y''(0) = C$	

$\boxed{} \rightarrow$ D.E $\rightarrow$ time domain $\xrightarrow{L.T}$ Freq $\xrightarrow{I.L.T}$ time domain.

Scanned with OKEN Scanner

1st order, Linear.
non-homogeneous ordinary differential equation.

Ex-1: Solve IVP, $\frac{dy}{dt} - 2y = e^{5t}$, $y(0) = 3$.

Soln: $\mathcal{L}\{\frac{dy}{dt}\} - 2\mathcal{L}\{y\} = \mathcal{L}\{e^{5t}\}$ | $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

$\Rightarrow sY(s) - y(0) - 2\cdot Y(s) = \frac{1}{s-5}$

$\Rightarrow sY(s) - 3 - 2Y(s) = \frac{1}{s-5}$

$\Rightarrow Y(s)(s-2) = \frac{1}{s-5} + 3 = \frac{1+3s-15}{s-5} = \frac{3s-14}{s-5}$

$\therefore Y(s) = \frac{3s-14}{(s-5)(s-2)}$. → Solution of frequency domain

1. Do Laplace
2. Input values of $y(0)$, $y'(0)$ etc
3. Free $Y(s)$.
4. I.L.T of $Y(s)$.

$\Rightarrow \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{3s-14}{(s-5)(s-2)}\right\}$

$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{3s-14}{(s-2)(s-5)}\right\}$.

$\therefore \frac{3s-14}{(s-2)(s-5)} = \frac{A}{s-2} + \frac{B}{s-5}$ [partial fraction].

$\therefore A = 8/3$, $B = 1/3$.

$\therefore y(t) = \mathcal{L}^{-1}\left\{\frac{8/3}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{1/3}{s-5}\right\}$.

$= 8/3\, \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + 1/3\, \mathcal{L}^{-1}\left\{\frac{1}{s-5}\right\}$

$$\boxed{y(t) = 8/3 \cdot e^{2t} + 1/3\, e^{5t}}$$ → Solution of time domain.

Ans:

Scanned with OKEN Scanner

Ex-2: $\frac{d^2y}{dt^2} + y = e^{-2t}\sin t$; $y(0) = 0$, $y'(0) = 0$.

Soln: $\mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} + \mathcal{L}\{y\} = \mathcal{L}\{e^{-2t}\sin t\}$.

$$\Rightarrow s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{1}{s^2+1}\Big|_{s \to s+2}$$

$$\Rightarrow s^2Y(s) + Y(s) = \frac{1}{(s+2)^2+1}$$

$$\Rightarrow Y(s)\,(s^2+1) = \frac{1}{(s+2)^2+1}$$

$$\Rightarrow Y(s) = \frac{1}{(s^2+1)\{(s+2)^2+1\}}$$

$$\therefore \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4s+5)}\right\}$$

Now,

$$\frac{1}{(s^2+1)(s^2+4s+5)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+4s+5}$$

$\therefore A + C = 0$; $4A + B + D = 0$; $5A + 4B + C = 0$, $5B + D = 1$.

$\therefore A = -1/8$, $B = 1/8$, $C = 1/8$, $D = 3/8$

$$\therefore \mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4s+5)}\right\} = \mathcal{L}^{-1}\left\{\frac{-1/8\,s + 1/8}{s^2+1} + \frac{1/8\,s + 3/8}{s^2+4s+5}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{-1/8\,s}{s^2+1}\right\} + \mathcal{L}^{-1}\left\{\frac{1/8}{s^2+1}\right\} + \mathcal{L}^{-1}\left\{\frac{1/8\,s}{s^2+4s+5}\right\} + \mathcal{L}^{-1}\left\{\frac{3/8}{s^2+4s+5}\right\}$$

$$= -1/8\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} + 1/8\,\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} + 1/8\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+5}\right\}$$

$$+ 3/8\,\mathcal{L}^{-1}\left\{\frac{1}{s^2+4s+5}\right\}.$$

Scanned with OKEN Scanner

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\{F(s+a)\}\cdot e^{at} \Rightarrow \text{shifting.}$$

$$= -\tfrac{1}{8}\cos t + \tfrac{1}{8}\sin t + \tfrac{1}{8}\mathcal{L}^{-1}\left\{\frac{s}{(s+2)^2+1}\right\} + \tfrac{3}{8}\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+1}\right\}.$$

shifting

$$y(t) = -\tfrac{1}{8}\cos t + \tfrac{1}{8}\sin t + \tfrac{1}{8}e^{-2t}\cos t + \tfrac{3}{8}e^{-2t}\sin t.$$

Ans: both (0, 0).

Ex-3: $y''(t) - 2y'(t) + y(t) = e^t$; $y(0) = 0$, $y'(1) = 1$.

boundary value

initial value $y(0)$, $y'(0) = 0$

boundary value $y(0) = 0$, $y'(0) = 1$.

Soln: $\mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} + \mathcal{L}\{y\} = \mathcal{L}\{e^t\}$

$$\Rightarrow s^2Y(s) - sy(0) - \boxed{y'(0)} - 2(sY(s) - y(0)) + Y(s) = \frac{1}{s-1}.$$

value not given.

$$\Rightarrow s^2Y(s) - 0 - a - 2(sY(s) - 0) + Y(s) = \frac{1}{s-1}.$$

$$\Rightarrow Y(s)(s^2 - 2s + 1) = \frac{1}{s-1} + a$$

$$\Rightarrow Y(s) = \frac{1}{(s-1)(s-1)^2} + \frac{a}{(s-1)^2}$$

$$\therefore Y(s) = \frac{1}{(s-1)^3} + \frac{a}{(s-1)^2}$$

$$\therefore \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^3}\right\} + \mathcal{L}^{-1}\left\{\frac{a}{(s-1)^2}\right\}.$$

$\mathcal{L}^{-1}\{F(s)\} = e^{at}\cdot\mathcal{L}^{-1}\{F(s+a)\}$

$\mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^3}\right\} + a\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^2}\right\} \rightarrow \text{shifting.}$$

$$\Rightarrow y(t) = \frac{1}{2!}\mathcal{L}^{-1}\left\{\frac{2!}{(s+1-1)^{2+1}}\right\}\cdot e^t + a\mathcal{L}^{-1}\left\{\frac{1!}{(s+1-1)^{1+1}}\right\}e^t$$

$$\boxed{y(t) = \frac{1}{2}t^2e^t + a\cdot te^t.}$$

Ans:

$$\therefore y'(t) = \frac{1}{2}(2te^t + t^2e^t) + a(e^t + te^t).$$

$$\therefore y'(1) = \frac{1}{2}(2\times 1\times e^1 + 1^2e^1) + a(e^1 + 1e^1) = \frac{1}{2}(2e + e) + a\cdot 2e$$

$$= \frac{3}{2}e + 2ae$$

Scanned with OKEN Scanner

$\Rightarrow 1 = \frac{3}{2}e + 2ae$

$\Rightarrow 1 = e\left(\frac{3}{2} + 2a\right)$

$\Rightarrow \frac{3}{2} + 2a = \frac{1}{e}$

$\Rightarrow a = \frac{\frac{1}{e} - \frac{3}{2}}{2} = \frac{\frac{2-3e}{2e}}{2} = \frac{2-3e}{4e} = \frac{1}{2e} - \frac{3}{4}$

$\therefore \boxed{y(t) = \frac{1}{2}t^2e^t + \left(\frac{1}{2e} - \frac{3}{4}\right)te^t.}$ Ans:

System of differential equation

Ex-4: $\frac{dx}{dt} - 6x + 3y = 8e^t$ $\quad x(0) = -1;\ y(0) = 0.$

$\frac{dy}{dt} - 2x - y = 4e^t$

Soln: 1st eqn

$\mathcal{L}\left\{\frac{dx}{dt}\right\} - 6\mathcal{L}\{x\} + 3\mathcal{L}\{y\} = 8\mathcal{L}\{e^t\}.$

$\Rightarrow sX(s) - x(0) - 6X(s) + 3Y(s) = 8 \cdot \frac{1}{s-1}$

$\Rightarrow sX(s) + 1 - 6X(s) + 3Y(s) = \frac{8}{s-1}$

$\Rightarrow X(s)(s-6) + 3Y(s) = \frac{8}{s-1} - 1 = \frac{-s+9}{s-1}$

$\therefore \boxed{(s-6)X(s) + 3Y(s) = \frac{-s+9}{s-1}}$ ——(i)

2nd eqn

$\mathcal{L}\left\{\frac{dy}{dt}\right\} - 2\mathcal{L}\{x\} - \mathcal{L}\{y\} = 4\mathcal{L}\{e^t\}$

$\Rightarrow sY(s) - y(0) - 2X(s) - Y(s) = 4 \cdot \frac{1}{s-1}$

$\Rightarrow \boxed{Y(s)(s-1) - 2X(s) = \frac{4}{s-1}}$ ——(ii)

Scanned with OKEN Scanner

(i) and (ii) :

$$(s-6)X(s) + 3Y(s) = \frac{-s+9}{s-1}$$

$$-2X(s) + (s-1)Y(s) = \frac{4}{s-1}$$

(i) × (s−1) − (ii) × 3 :

$$(s-6)(s-1)X(s) + 3(s-1)Y(s) = -s+9$$

$$-2\cdot 3X(s) + 3(s-1)Y(s) = \frac{12}{s-1}$$

(+) (−) (−)

$$(s-6)(s-1)X(s) + 6X(s) = -s+9 - \frac{12}{s-1}$$

$$\Rightarrow X(s)\left((s-6)(s-1)+6\right) = \frac{-s^2+s+9s-9-12}{s-1}$$

$$\Rightarrow X(s) = \frac{-s+7}{(s-1)(s-4)}.$$

$$Y(s) = \frac{2}{(s-1)(s-4)}.$$

$$\therefore \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{-s+7}{(s-1)(s-4)}\right\}$$

$$\Rightarrow \boxed{x(t) = -2e^{t} + e^{4t}}$$

$$\therefore \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{(s-1)(s-4)}\right\}$$

$$\Rightarrow \boxed{y(t) = -\frac{2}{3}e^{t} + \frac{2}{3}e^{4t}.}$$

Ans:

$$X(s) = \frac{s^2+10s-21}{((s-6)(s-1)+6)\,(s-1)}$$

Scanned with OKEN Scanner